

Current Procedure

This procedure will allow you to prepare an STL file of an amino acid side chain. It does not currently include the steps to print the finished file.

ChemDraw:

1. Open **ChemDraw**.
2. Select **File -> Open Templates -> Amino Acid Side Chains**.
3. Select the side chain of the amino acid you would like to print. They are organized in a table by their three-letter abbreviation.
4. Under the tools bar, select **Lasso**.
5. Use the lasso tool to highlight the amino acid side chain.
6. Copy the highlighted side chain using **CTRL+C** or **right click -> Copy**. You will paste the amino acid structure directly into Chem3D.

Chem3D:

7. Open **Chem3D**.
8. On the right side under "ChemDraw - LiveLink" paste the side chain using **CTRL+V** or **right click -> Paste**.
9. In the top bar, select **Calculations -> MM2 -> Minimize Energy**.
10. Select **Run**.
11. Select **File -> Save As**.
12. Save your file with "**SYBYL2 (*.mol2)**" as the file type.

Chimera:

13. Open **Chimera**.
14. Select **File -> Open**.
15. Select your .mol2 file.
16. Select **Select -> Chemistry -> Element -> H**. (Skip if you want to show hydrogen in your model).
17. Select **Actions -> Atoms/Bonds -> Hide**. (Skip if you want to show hydrogen in your model).
18. Select **Favorites -> Command Line**.
19. Copy the following into the command line located at the bottom of the app:

```
represent b+s  
~disp solvent  
setattr M ballScale 0.3
```

```
setattr M stickScale 1.2  
bondcolor grey  
setattr p drawMode 1
```

20. Select **File** -> **Export Scene**

21. Save your file with “**VMRL [.wrl, .vmrl]**” as the file type.

Blender:

22. Open **Blender**.

23. Click “x” to delete the cube that is in the file by default.

24. In the side bar, select **MolPrint**.

25. Select **Open VMRL**.

26. Select your .vmrl or .wrl file.

27. Select **Open VMRL2**.

28. Input the following settings (if there are double bonds):

double bond scale (DB scale) = 0.40

double bonds separation (DB separ) = 1.41

29. If there are any double bonds in the amino acid, right click where the double bond should be and select **Double Bonds** on the left side.

30. On the carbon that attaches to the backbone, any attached hydrogen bonds or atoms.

31. Select **File** -> **Export** -> **Stl (.stl)**.

32. Save your file.